



COURSE SYLLABUS						
Course Code	INTE 303	Course Title	Capstone Project 1		Course Credit	3
Course Description		Capstone Project 1 serves as the initial phase of the students’ culminating requirement in the BS Information Technology program. This course focuses on the conceptualization, research, and proposal development of an IT-based solution that addresses a real-world problem. Students will identify a research problem, conduct a comprehensive review of related literature, design appropriate methodologies, and prepare system analysis and design documentation. The course emphasizes critical thinking, collaboration, and adherence to research and ethical standards as students prepare for their proposal defense. By the end of the course, students are expected to produce and defend a complete Capstone Project proposal (Chapters 1–3).				
Pre-Requisites		COMP 015	Co-Requisites	None		
VISION		A Leading Comprehensive Polytechnic University in Asia				
MISSION		Advance an inclusive, equitable, and globally relevant polytechnic education towards national development.				
QUALITY POLICY STATEMENT		The Polytechnic University of the Philippines commits to provide inclusive and equitable quality education and promote lifelong learning opportunities for human advancement and social transformation through re-engineered polytechnic academic programs. Toward this end, we, the members of the PUP Community will vigorously and steadfastly endeavor to continuously improve the standard of University services to the satisfaction of our clients through the adoption and continuous review of our Quality Management System.				



**INSTITUTIONAL LEARNING
OUTCOMES
(ILO)**

As a polytechnic state university, PUP shall develop its students to possess:

1. **Critical and Creative Thinking.** Graduates use their rational and reflective thinking as well as innovative abilities to life situations in order to push boundaries, realize possibilities, and deepen their interdisciplinary, multidisciplinary, and/or transdisciplinary understanding of the world.
2. **Effective Communication.** Graduates apply the four macro skills in communication (reading, writing, listening, and speaking), through conventional and digital means, and are able to use these skills in solving problems, making decisions, and articulating thoughts when engaging with people in various circumstances.
3. **Strong Service Orientation.** Graduates exemplify strong commitment to service excellence for the people, the clientele, industry, and other sectors.
4. **Adept and Responsible Use or Development of Technology.** Graduates demonstrate optimized and responsible use of state- of-the-art technologies of their profession. They possess digital learning abilities, including technical, numerical, and/or technopreneurial skills.
5. **Passion for Lifelong Learning.** Graduates perform and function in society by taking responsibility in their quest for further improvement through lifelong learning.

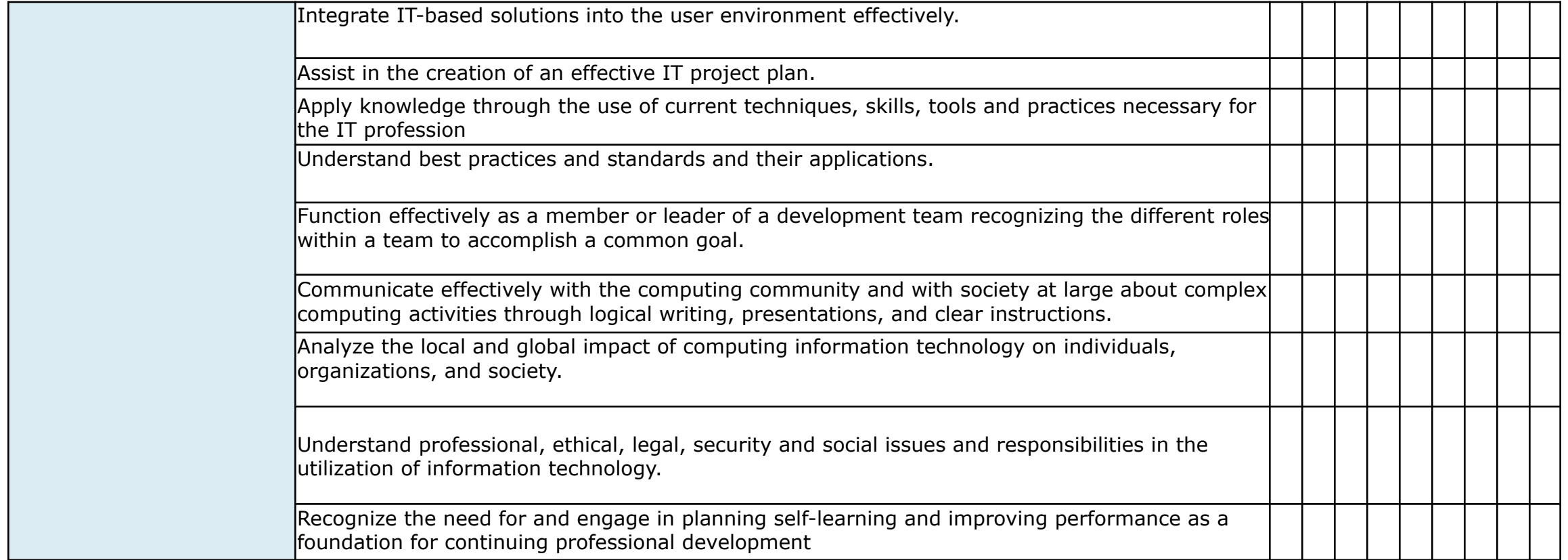


	6. Leadership and Organizational Skills. Graduates assume leadership roles and become leading professionals in their respective disciplines by equipping them with appropriate organizational skills. 7. Personal and Professional Ethics. Graduates manifest integrity and adherence to moral and ethical principles in their personal and professional circumstances. 8. Resilience and Agility. Graduates demonstrate flexibility and the growth mindset to adapt and thrive in the volatile, uncertain, complex and ambiguous (VUCA) environment. 9. National and Global Responsiveness. Graduates exhibit a deep sense of nationalism as it complements the need to live as part of the global community where diversity is respected. They promote and fulfill various advocacies for human and social development.
COLLEGE GOALS AND OBJECTIVES	The Polytechnic University of the Philippines – Taguig Campus aims to achieve the following goals: 1. Offer curricula which are relevant and responsive to the changing needs of the industry and society; 2. Provide trainings to students that will enable them to become potent instruments to socio-economic development. 3. Prepare students to become productive, creative, innovative, and dynamic in their chosen fields of specialization. 4. Provide state-of-the-arts facilities of learning to optimize student development; 5. Inculcate appropriate values that are necessary to build a humane, disciplined, nationalist, and independent society; 6. Tap potentials of students, faculty, administrative staff and other stakeholders in formulating policies for institutional development; 7. Involve all stakeholders in enhancing programs for student’s faculty and employees’ development; 8. Partner with industry and other research institutions strengthening research capabilities of faculty, employees and students; 9. Facilitate presentation of research outputs in international fora, and their publication in recognized journals; 10.Undertake outreach and research-based extension programs by tapping the expertise and other resources of all stakeholders and, 11.Produce technologies for commercialization of livelihood improvement.

PROGRAM LEARNING OUTCOMES (PLO)	Based on CMO	Alignment to ILOs								
		1	2	3	4	5	6	7	8	9
		Apply knowledge of computing, science, and mathematics appropriate to the discipline.								
		Analyze complex problems, and identify and define the computing requirements appropriate to its solution								
	Identify and analyze user needs and take them into account in the selection, creation,									



	evaluation and administration of computer-based systems.											
	Design, implement, and evaluate computer-based systems, processes, components, or programs to meet desired needs and requirements under various constraints.											



COURSE LEARNING OUTCOMES (CLO)		Alignment to PLOs							
		1	2	3	4	5	6	7	8
	Identify and define a real-world problem that can be addressed through an IT-based solution.								
	Conduct background research and needs analysis to establish project significance and feasibility.								
	Formulate clear project objectives, scope, and requirements aligned with user and organizational needs.								

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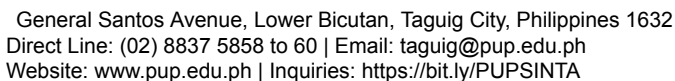
	Develop a comprehensive review of related literature and systems to support the proposed project.								
	Apply appropriate research methodologies and data-gathering techniques relevant to the project.								
	Design the system architecture, process models, and database schema for the proposed system.								
	Prepare a well-structured capstone project proposal document following institutional guidelines.								
	Present and defend the proposed project before a panel of evaluators with clarity and professionalism.								





OUTCOMES-BASED TEACHING AND LEARNING PLAN (OBTL PLAN)

Week	Desired Learning Outcomes (DLOs)	Learning Content/ Topics	Instructional Delivery Design		Assessment		Alignm ent to CLOs			
			Face-to-Face	Flexible Teaching and Learning Activities (FLTAs)			1	2	3	4
					Face-to-Face	FLTAs				
1	- Equip students on the pivotal ideas embedded in the vision, mission, and 10 Pillars of PUP - Gain an overview of the course subject and requirements	- Orientation about PUP Vision, Mission, and 10 Pillars - Overview about the Subject Course and Course Requirements	Present the VMGO, policies, and guidelines on the conduct of Face-to-Face; and of the requirements of the course.		Formative Assessment (Sharing of Learning)		X			
2	Understand the course objectives, requirements, and stages of Capstone Project 1; Identify researchable IT project problems aligned with institutional or community needs.	Course orientation and Capstone 1 overview; Identifying research problems and project opportunities; Group formation.	Orientation and brainstorming sessions; discussion of Capstone guidelines; topic consultation (synchronous and asynchronous).	- Online discussion of Capstone guidelines. - Brainstorming of project ideas. - Topic consultation via LMS.	Participation during orientation and consultations.	Submission of list of possible project topics.				
3	Formulate clear	Chapter 1:	Lecture and peer review.	Guided online writing of	Review and critique of	Submission of Chapter 1 draft.				





				Online Q&A session for tool selection.					
8	Design and prepare data gathering instruments and define detailed system requirements based on user analysis.	Instruments, Data Collection, and Sampling Procedures; Requirements Analysis and Specification.	Workshop on survey/interview design and requirements gathering.	- Virtual workshop on survey/interview design. - Consultation for requirements documentation.	Checking of survey and interview tools.	Submission of validated instrument and requirements document.			
9	MIDTERM WRITTEN EXAMINATION								
10	Create initial system design models.	Use Case Diagram, DFD, and ERD.	Laboratory design session.	- Online demo of diagram tools. - Continued system modeling.	Draft of system models.	- Synchronous review of diagrams - Upload of system models file			
11	Prepare project timeline and development plan.	Operational, Technical, Economic, and Schedule Feasibility.	Discussion and case examples.	- Real-time consultation. - Feasibility writing and computation.	Feasibility study draft.	- Online feedback on feasibility computations - Upload of feasibility study			



						draft				
12	Integrate Chapters 1–3 into a cohesive proposal.	Integration and refinement of Capstone Document.	Instructor-guided peer review.	• Online proofreading workshop. • Editing and document polishing.	Draft of Chapters 1–3.	• Group discussion and critique session • Submission of full Chapters 1–3 draft				
13	Present preliminary proposal and gather feedback.	Proposal Presentation Techniques and Guidelines.	Mock proposal presentation.	• Presentation rehearsal online. • Review and improvement.	Pre-oral presentation.	• Online rehearsal and Q&A • Upload of presentation slides and script				
14	Revise project proposal based on feedback.	Revision process and consultation.	One-on-one faculty consultation.	• Q&A for revisions. • Revision and rewriting.	Revised Capstone Proposal.	• Virtual consultation for improvements • Resubmission of revised chapters				
15	Prepare necessary documentation for oral defense.	Formatting, printing, and compliance checking.	Document formatting workshop.	• Review of submission requirements. • Final edit and compilation.	Finalized Capstone Paper for defense.	• Online pre-defense checklist review • Upload of final				



						formatted manuscript				
16	Deliver oral presentation effectively.	Proposal Defense proper.	Panel defense and Q&A.	• None (face-to-face defense week). • Review notes for improvement.	Oral defense and grading.	• Reflection post-defense via online form				
17	• Apply feedback and finalize approved project proposal. • Reflect on the capstone research process and future direction.	• Incorporating panel recommendations. • Reflection and documentation of learning.	• Consultation for revisions. • Reflection discussion and evaluation.	• Follow-up consultation online. • Revision and resubmission. • Closing synchronous reflection. • Closing synchronous reflection.	• Final approved manuscript. • Reflection paper / Course evaluation.	• Virtual consultation for final edits • Upload of approved final proposal • Online exit interview and reflection sharing • Submission of individual reflection paper and self-assessment				
18	FINAL EXAMINATION ()									



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
OFFICE OF THE VICE PRESIDENT FOR CAMPUSES
TAGUIG CAMPUS
Bachelor of Science in Information Technology
(BSIT)



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		Resources and Applying Strategies in Teaching Prose <i>Competency 4:</i> Recognizing Assessment in Teaching Prose	achievement which are consistent with the selected competencies • Learning Plan Making according to the English curricula that develops higher order thinking skills of learners through the use of prose and dramatic texts		• Quizzes					
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FINAL EXAMINATION

LIST OF REFERENCES

BOOKS:

WEBSITE:

- Lucidchart. (2023, November 8). How to create a conceptual framework for research. Lucidchart Blog. <https://www.lucidchart.com/blog/how-to-make-a-conceptual-framework>
- Visual Paradigm. (2023, May 12). How to create data flow diagrams (DFD): A beginner’s guide. Visual Paradigm. <https://www.visual-paradigm.com/guide/data-flow-diagram/what-is-data-flow-diagram/>
- IBM. (2024, February 15). What is system analysis and design? IBM Developer. <https://developer.ibm.com/articles/system-analysis-design/>
- Project Management Institute. (2023, October 2). What is a feasibility study and why is it important? PMI Knowledge Center. <https://www.pmi.org/learning/library/feasibility-study-importance-12345>
- ResearchGate. (2024, January 9). Best practices in writing a research methodology for IT capstone projects. ResearchGate. <https://www.researchgate.net/publication/IT-capstone-methodology-best-practices>



CLASSROOM POLICY	
FACE-TO-FACE DELIVERY	FLEXIBLE TEACHING AND LEARNING ACTIVITIES (FLTAs)
Please refer to the FLTA policies, except for the stipulation on Attendance. Attendance will be checked during face-to-face sessions.	Students’ Responsibilities Students are expected to: a. Take time to read and watch the provided class learning materials. b. Fulfill class requirements such as posting on discussion boards and submitting group presentations. c. Check the status of their submissions and inform the instructor if they are having problems submitting the requirements.



The instructor may not be able to remind all students about their missing/unsubmitted requirements. If at the end of the course students still have missing requirements, the instructor will compute for the final grade without those submissions (They will be graded with 0) and submit the grade. If the final grade is "INC" grade, the student must contact the instructor to decide in completing the course, ASAP.

d. Confer with classmates or the instructor about the discussions during the synchronous sessions that they missed if unable to attend the live sessions. Recordings will be provided but may not be available immediately.

e. Inform the instructor of concerns that may prevent them from fully participating in the class. This includes withdrawal from the course, long term illnesses, personal matters, and others. Teachers will not be able to help you if you don't tell them of these concerns.

Attendance.
We recognize that students have varying access levels to technology and connectivity, which is why for the most part, the course is conducted asynchronously. During synchronous sessions, there will be no checking of attendance. However, everyone is encouraged to participate in these sessions. A recording and/or transcript of synchronous classes will be made available for students who are unable to attend a synchronous session.

Late Submissions.
Deadlines will be set for the submission of course requirements. This allows the instructor sufficient time to review them. It also helps students pace their work. If you experience problems submitting your work on time, please contact the instructor ahead of time. All requirements must be completed to pass the course. Late submissions will not merit deductions but may not receive timely feedback.

Cheating and Plagiarism.
Any work submitted by a student in this course for academic credit will be the student's own work. Any form of academic dishonesty will be fully dealt with according to the stipulations of the PUP Student Handbook.

Participation.
Participation during discussions is greatly encouraged, and students should feel free to ask questions when an idea or lesson is unclear or confusing, especially during synchronous sessions.

Announcements.
Class announcements will be communicated through the class president. Other communication shall always be coursed through university-issued (i.e., official) email accounts. Students must use their PUP student accounts.

Intellectual Property and Copyright.
Original materials created for this online class is covered by university policies on intellectual property. Some (third-party) instructional materials (e.g., YouTube videos) might be under the public domain or are open access, while others (e.g., some assigned readings) might be subject to fair use. All third-party instructional materials will



	be attributed to their original creators. Electronic copies of the assigned readings are for your private consumption only. Access to the course materials for this class for PUP students will be restricted to only those students registered in this course and will be available for only a
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	limited period. Please do not share these materials with anyone elsewhere. The same applies to the recordings of the synchronous sessions.	
COURSE REQUIREMENT/S	GRADING SYSTEM	
Students are required to prepare and submit a comprehensive Capstone Project Proposal covering Chapters 1 to 3. They must develop system analysis and design artifacts, including use case diagrams, data flow diagrams, entity-relationship diagrams, and a conceptual framework. A feasibility study assessing the project’s technical, operational, and economic viability must be completed. Students are expected to attend regular consultations and submit progress reports to demonstrate consistent development. They must also present and defend their proposal before a panel and submit a final revised and approved version of their project proposal.	Mid Term	Final Term
	Class Standing 70% Exercises/Activities Summative Quizzes Recitation and Attendance Reflection Paper	Class Standing 70% Exercises/Activities Summative Quizzes Recitation and Attendance Reflection Paper
	Midterm Examinations 30%	Final Examinations 30%
	100%	100%
	Final Grade = (Midterm Grade + Final Term Grade) / 2	
CLASS INFORMATION		FACULTY INFORMATION
Course, Year and Section: BSIT 3-1		Name of Faculty: Gecilie C. Almirañez
Day and Time: Tuesday 10:00 AM-3:00PM		Consultation Time:
Room: TGDOST		Office Tel. No./ Mobile Phone No.



School Year: 2526	Semester: 2nd Semester	E-Mail Address:
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Prepared by:	Reviewed and checked by:	Recommending Approval:	Approved by:
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